

Economic Growth, the Financial System, and Business Cycles

ECON 101 – Macroeconomics

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Chapter Outline

- **6.1** Long-Run Economic Growth
- 6.2 Saving, Investment, and the Financial System
- 6.3 The Business Cycle

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6.1 Learning Objective

Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising **productivity** increases the **average standard of living**.
- This differs from short-run fluctuations (**business cycles**):
 - Expansion: real GDP rising.
 - Recession: real GDP falling.
- The most common measure of the standard of living:
 - $\text{Real GDP per capita} = \text{real GDP} \div \text{population}$.

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Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
 - How much goods and services, on average, each person can consume.
 - Adjusted for changes in the price level (**real**, not nominal).
- Countries with higher real GDP per capita typically have:
 - better health, education, housing, and infrastructure.
- **Canadian context:** In 2023, Canada's real GDP per capita was approximately **\$53,000** (2012 constant dollars), roughly **four times** the 1961 level of $\approx \$13,800$.
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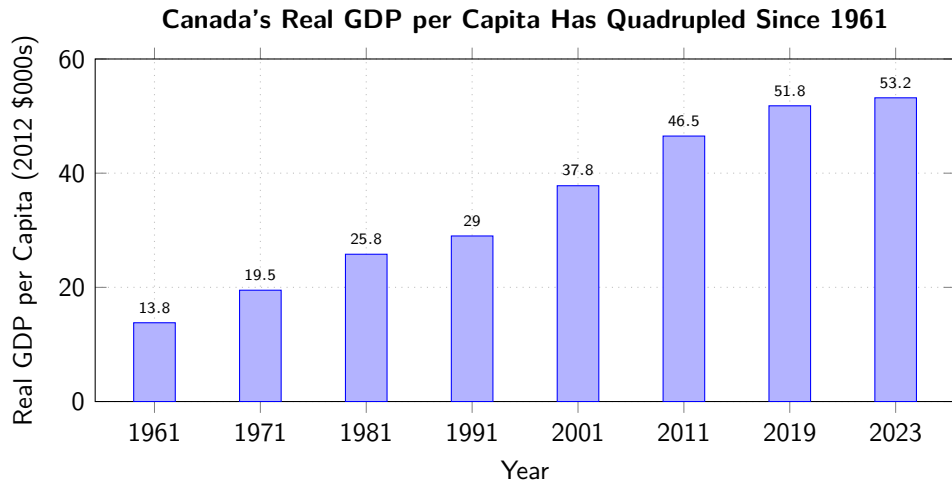
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Figure 6.1 – Real GDP per Capita in Canada, Selected Years



Source: Statistics Canada, Table 36-10-0222-01 (2012 constant dollars).

Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita has risen **nearly four-fold** since 1961.
- The average Canadian in 2023 commands **almost four times as many** goods and services as the average Canadian in 1961.
- This long-run rise is what we call **economic growth**.
- **Notable interruptions:**
 - Early 1980s recession (high interest rates, oil price shock).
 - Early 1990s recession (tight monetary policy, housing bust).
 - 2008–2009 Global Financial Crisis.
 - **2020 COVID-19 pandemic:** real GDP per capita fell $\approx 13\%$ in just 2 quarters — the sharpest decline on record.

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Calculating Growth Rates

- Growth rate of an economic variable from year $t - 1$ to year t :

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, compute growth each year and take the average.
- **Canadian example:**
 - Real GDP per capita in 2022: \$53,100.
 - Real GDP per capita in 2023: \$53,200.
 - Growth rate $\approx \frac{53,200 - 53,100}{53,100} \times 100 \approx 0.19\%$.
 - Canada's 2023 growth was very weak — immigration drove population faster than real output.

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The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- **Canadian examples:**
 - Canada's historical average growth of real GDP per capita $\approx 2\%$:
 - Years to double $\approx 70/2 = 35$ years.
 - At only 1% growth: $70/1 = 70$ years — twice as long!
- Small differences in growth rates **compound** dramatically over decades.

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Rule of 70: Country Comparison

Country A — 2% growth

- Years to double: $70/2 = 35$
- Income after 35 years: **2x**
- Income after 70 years: **4x**

Country B — 4% growth

- Years to double: $70/4 = 17.5$
- Income after 35 years: **4x**
- Income after 70 years: **16x**

- **Real example:** South Korea's economy grew $\approx 7\%$ per year from 1960 to 2000, doubling every 10 years — raising per capita income from $\approx \$1,000$ to $\approx \$20,000$ (2000 US\$).
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What Determines Long-Run Growth?

- Long-run increases in real GDP per capita depend on **labour productivity**:

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Number of workers (or hours worked)}}.$$

- The average Canadian produces much more per hour than in 1961.
- Two key drivers:
 1. Increases in capital per hour worked
 2. Technological change
- **Canadian data (Statistics Canada):** Labour productivity grew an average of 1.4% per year from 2000–2023 — slower than the G7 average, raising policy concern.

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Capital per Hour Worked

- **Physical capital:** machines, computers, tools, buildings, vehicles.
 - More capital per worker $\Rightarrow$ more output per hour.
- **Human capital:**
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- **Canadian example:**
 - A Canadian oil sands worker equipped with modern extraction technology and earth-moving equipment produces orders of magnitude more output per day than a 1961 worker with manual tools.
 - BC's tech sector: engineers with university degrees and high-powered workstations represent both physical and human capital investment.

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- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
 - New production processes (AI-assisted manufacturing).
 - Better management (lean production, just-in-time inventory).
 - New products (mRNA vaccines, electric vehicles).
- Workers produce more **with the same inputs** — an outward shift of the production frontier.
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Potential GDP

- **Potential GDP:** the level of real GDP attained when all firms produce at **capacity** —
 - “normal” hours, “normal” workforce, no emergency overtime.
- Also called **full-employment GDP** or **trend GDP**.
- Potential GDP grows when:
 - the labour force expands (e.g., through immigration),
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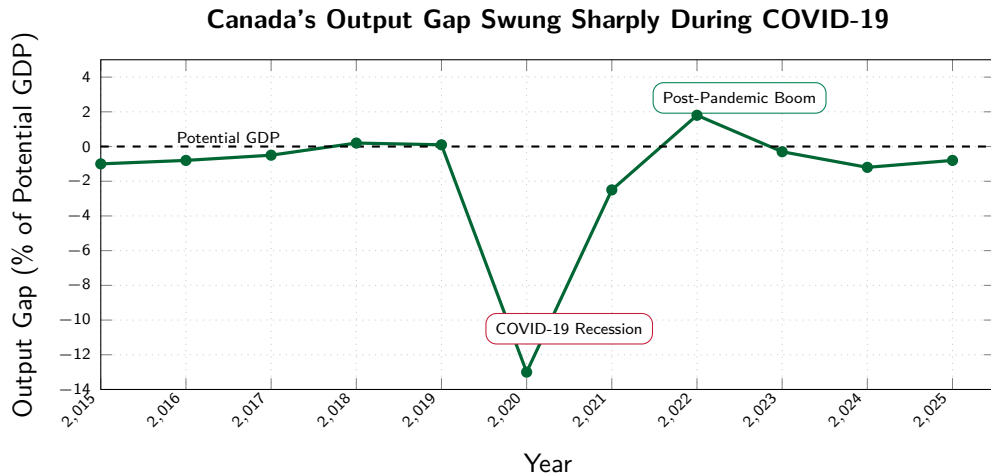
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Figure 6.2 – Output Gap: Canada 2015–2025 (Estimated)



Source: Bank of Canada, Monetary Policy Report (estimates). Shaded area = negative output gap (recessionary).

The Output Gap

- **Output gap:** = actual GDP – potential GDP.
- When the output gap is **positive** (actual > potential):
 - economy is using resources **unsustainably** — very low unemployment, high capacity utilization, **inflationary pressure**.
 - **Canada 2022:** output gap $\approx +1.8\%$ coincided with CPI inflation peaking at **8.1%**.
- When the output gap is **negative** (actual < potential):
 - resources (labour, capital) are **underutilized**.
 - **Canada 2020:** output gap plunged to $\approx -13\%$ during COVID-19 lockdowns.
- The Bank of Canada uses the output gap to guide its monetary policy decisions (MPR, 2024).

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6.2 Learning Objective

Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
 - more capital per worker, and ongoing technological improvements.
- Both require **investment**.
- Investment is financed by **saving**, channeled through the financial system.
- **Canadian saving rate** (household, % of disposable income): fell from $\approx 18\%$ in 1982 to $\approx 2\text{--}4\%$ pre-COVID, then **surged** to 28% during 2020 lockdowns (Statistics Canada).

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Three Key Services of the Financial System

Risk-Sharing

- Spread money across many assets.
- Reduces risk while maintaining returns.
- *E.g., RBC Select Balanced mutual fund.*

Liquidity

- Ease of converting to cash.
- Bank accounts, money market funds, publicly traded stocks.
- Allows long-term commitments with short-term access.

Information

- Prices of securities aggregate expectations.
- Guide funds toward the most productive uses.
- Reduce adverse selection and moral hazard.

The Macroeconomics of Saving and Investment

- National income identity:

$$Y = C + I + G + NX.$$

- For simplicity, assume a **closed economy** ($NX = 0$):

$$Y = C + I + G \Rightarrow I = Y - C - G.$$

- Investment equals total income not consumed or used by government.
- **Private saving:**

$$S_{\text{private}} = Y + TR - C - T.$$

- **Public saving:**

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- If $S_{\text{public}} < 0$: government runs a **budget deficit**.
 - Government must borrow (sell bonds), reducing funds available for private investment.
- **Crowding out**: the reduction in private investment caused by government borrowing.
- **Canadian context**:
 - Federal government ran deficits every year 2009–2024.
 - 2020–21 deficit: \$314 billion (largest ever), due to COVID-19 emergency spending.
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- In practice, crowding-out effects on interest rates are partially offset by global capital flows.

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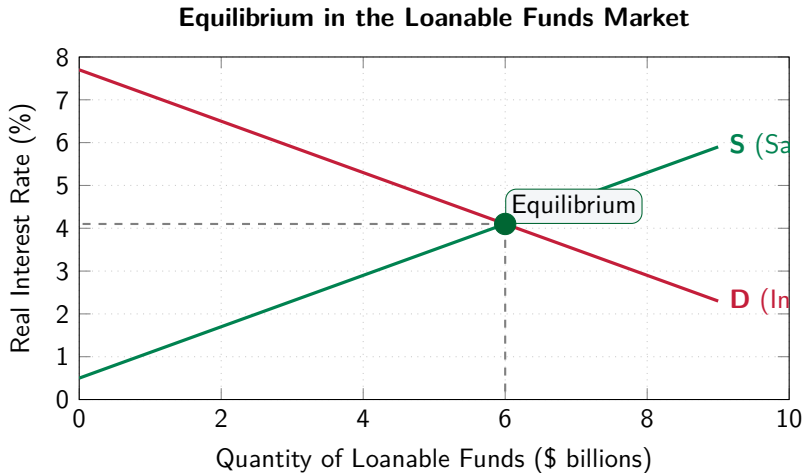
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Figure 6.3 – The Market for Loanable Funds



Supply and Demand for Loanable Funds

- **Demand for loanable funds (firms):**
 - Firms borrow to finance investment projects.
 - Borrow more when real interest rate is **lower** (more projects profitable).
 - **Canada 2020:** pandemic uncertainty slashed private investment by $\approx 15\%$.
- **Supply of loanable funds (households):**
 - Households save more when real interest rate is **higher** (greater reward to saving).
 - Government saving (surplus) adds to supply; deficits reduce supply.
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Supply and Demand for Loanable Funds

- **Demand for loanable funds** (firms):
 - Firms borrow to finance investment projects.
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 - **Canada 2020**: pandemic uncertainty slashed private investment by $\approx 15\%$.
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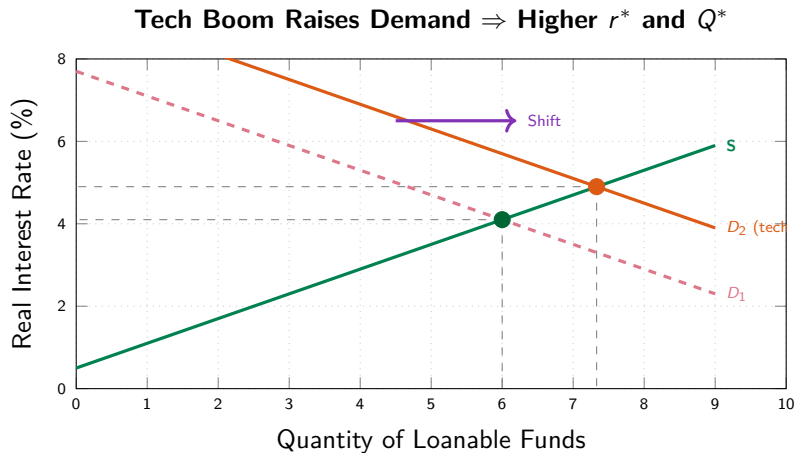
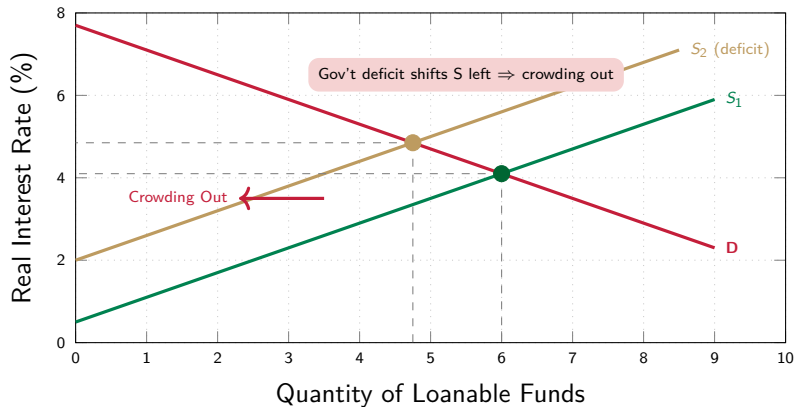


Figure 6.5 – Budget Deficit Crowds Out Investment

Government Deficit Reduces Supply $\Rightarrow$ Higher r^* , Lower Private Q^*



6.3 Learning Objective

Goal

Explain what happens during the business cycle and how it affects inflation and unemployment.

- Real GDP per capita has risen about **four-fold** since 1961, but not smoothly.
- Canada has experienced several recessions:
 - 1981–82, 1990–91, 2008–09, and 2020.
- The **business cycle** describes these short-run fluctuations.

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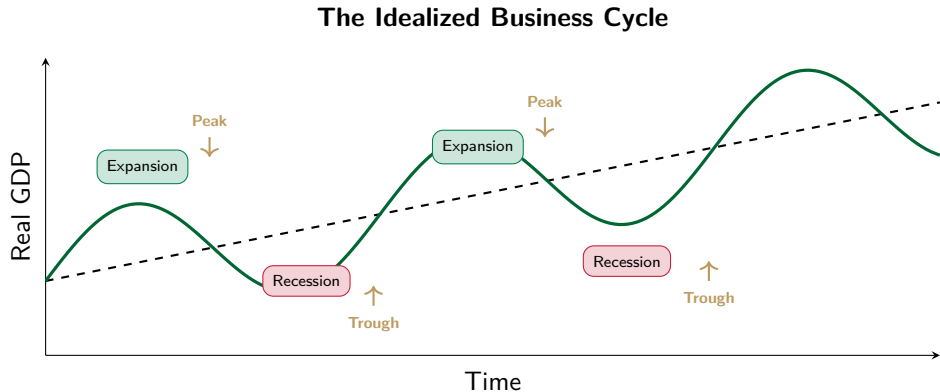
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Figure 6.6 – Phases of the Business Cycle



Phases of the Business Cycle

- **Expansion:** real GDP is rising (employment grows, incomes rise).
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling (often defined as 2+ consecutive quarters).
- **Trough:** the point where real GDP stops falling and begins to rise.
- A new expansion follows the trough.

Canadian Recessions Since 1980

Recession	Cause	Peak Unemployment
1981–82	Volcker rate shock	13.0%
1990–91	Tight monetary policy	11.9%
2008–09	Global financial crisis	8.7%
2020	COVID-19 pandemic	14.6% (May 2020)

Source: Statistics Canada, Labour Force Survey.

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How Does Canada Identify Recessions?

- There is **no official body** in Canada that declares recessions.
- Common rule of thumb:
 - Two consecutive quarters of negative real GDP growth.
- Statistics Canada (CANSIM) provides real GDP data quarterly and monthly.
- **2020 example:**
 - Q1 2020: real GDP fell -8.8% (annualized).
 - Q2 2020: real GDP fell -38.7% (annualized) — deepest quarterly drop on record.
 - Economy recovered to 2019 levels only by end of 2021.

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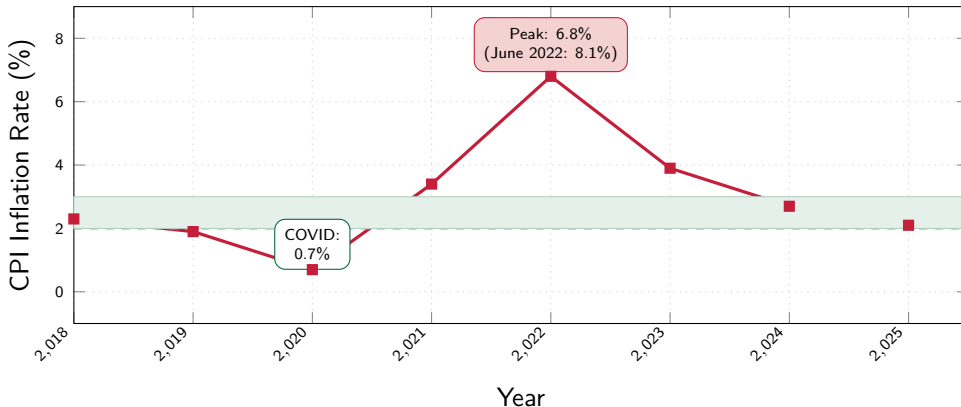
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Figure 6.7 – Inflation and the Business Cycle (Canada)

Canada CPI Inflation Fell in Recession, Surged in Recovery



Source: Statistics Canada, Table 18-10-0004-01. 2025 is estimated. BoC target band 1–3% shaded.

The Effect of the Business Cycle on Inflation

- **During recessions:**

- Demand falls below supply capacity.
- Firms compete for fewer customers $\Rightarrow$ prices rise more slowly or fall.
- **Canada 2020:** CPI inflation dropped to 0.7%.

- **During expansions:**

- Demand exceeds supply capacity.
- Firms can raise prices more easily $\Rightarrow$ higher inflation.
- **Canada 2022:** Post-COVID surge in demand + supply chain disruptions $\Rightarrow$ CPI peaked at 8.1% in June 2022 (Statistics Canada).

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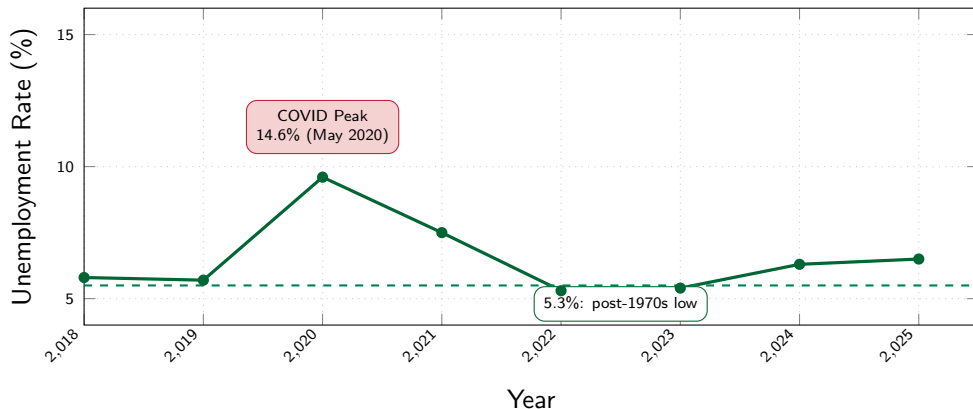
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Figure 6.8 – Unemployment and the Business Cycle (Canada)

Unemployment Rose Sharply in 2020, Then Gradually Climbed Again



Source: Statistics Canada, Labour Force Survey (Table 14-10-0017-01). 2025 is estimated.

The Effect of the Business Cycle on Unemployment

- **During recessions:**

- Firms see sales fall, reduce production, lay off workers.
- Unemployment often **continues to rise** even after the recession begins (lagging indicator).
- **Canada 2020:** unemployment hit **14.6%** in May 2020.

- **During expansions:**

- Firms hire more; unemployment gradually falls.
- **Canada 2022:** unemployment reached **5.3%**, near a 50-year low, amid very tight labour markets.

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Summary

- **Long-run economic growth** is driven by **labour productivity**, which depends on:
 - capital per worker, human capital, and technological change.
- Canada's real GDP per capita quadrupled since 1961 — mostly due to technology and capital deepening.
- **Potential GDP** and the **output gap** reveal whether resources are over- or under-utilized.
- The **financial system** (banks, markets, intermediaries) channels saving into investment via:
 - risk-sharing, liquidity, and information.
- In a closed economy, $S = I$; government deficits can **crowd out** private investment.
- The **business cycle** — expansions, peaks, recessions, troughs — is accompanied by:
 - higher inflation in booms, lower in recessions.
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- The **financial system** (banks, markets, intermediaries) channels saving into investment via:
 - risk-sharing, liquidity, and information.
- In a closed economy, $S = I$; government deficits can **crowd out** private investment.
- The **business cycle** — expansions, peaks, recessions, troughs — is accompanied by:
 - higher inflation in booms, lower in recessions.
 - rising unemployment in recessions, falling in expansions.
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Thank you!

Questions?